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Object Oriented Programming Using Java

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Objectives

- Operators in java.

Java Operators

- Divided into four categories.
  - Arithmetic
  - Bitwise
  - Relational
  - Logical

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Arithmetic Operators

Operator	Result
+	Addition
-	Subtraction
*	Multiplication
/	Division
%	Modulus
++	Increment
+=	Addition Assignment
--	Decrement
-=	Subtraction Assignment
*=	Multiplication Assignment
/=	Division Assignment

Bitwise Operators

Operator	Result
~	Bitwise unary NOT
&	Bitwise AND
|	Bitwise OR
^	Bitwise exclusive OR
>>	Shift right
<<	Shift Left
>>>	Shift right zero fill
&=	Bitwise AND assignment
|=	Bitwise OR assignment
^=	Bitwise exclusive OR assignment
>>=	Shift right assignment

Bitwise Logical Operators

A	B	A|B	A&B	A^B	~A
0	0	0	0	0	1
1	0	1	0	1	0
0	1	1	0	1	1
1	1	1	1	0	0

Shift Operators

- The right shift operator, >>, shifts all of the bits in a value to the right a specified number of times.

  value >> num

Eg. int a = 35; a = a >> 2; a = 8

- The left shift operator, << shifts all of the bits in a value to the left a specified number of times.

  value << num

Relational Operators

- Outcome is a Boolean value.

Operator	Result
==	Equal to
!=	Not equal to
>	Greater than
<	Less than
>=	Greater than or equal to
<=	Less than or equal to

Other Operators

Assignment Operators

- var = expression;
- The type of var must be compatible with the type of expression.

? Operator

- Java includes a special ternary (three-way) operator that can replace certain types of if-then-else statements
- expression1 ? expression2 : expression3

Takeaways

- Operators in Java.

Reference

[1] Herbert Schildt, *"Java: The Complete Reference, Eleventh Edition", Oracle 2018*